

Trainings ▾

Preparatory Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery first and connect the negative (-) battery last.

WARNING

Coolant is toxic; Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

WARNING

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

WARNING

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

WARNING

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

WARNING

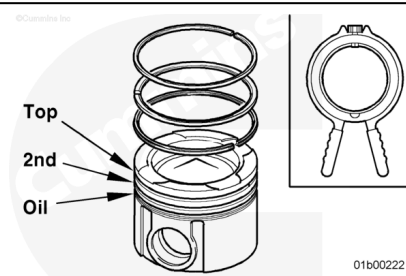
This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.

- Disconnect the battery or supply air to the starter.
- Drain the coolant. Refer to Procedure 008-018 in Section 8 (</qs3/pubsys2/xml/en/procedures/89/89-008-018.html>).

- Drain the lubricating oil and remove the oil pan. Refer to Procedure 007-025 in Section 7 (</qs3/pubsys2/xml/en/procedures/89/89-007-025.html>).
- Remove the oil suction tube and block stiffener plate. Refer to Procedure 001-089 in Section 1 (</qs3/pubsys2/xml/en/procedures/89/89-001-089.html>).
- Remove the intake manifold. Refer to Procedure 010-023 in Section 10 (</qs3/pubsys2/xml/en/procedures/89/89-010-023.html>).
- Remove the exhaust manifold. Refer to Procedure 011-007 in Section 11 (</qs3/pubsys2/xml/en/procedures/89/89-011-007.html>).
- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3 (</qs3/pubsys2/xml/en/procedures/89/89-003-011.html>).
- Remove the rocker lever assembly. Refer to Procedure 003-009 in Section 3 (</qs3/pubsys2/xml/en/procedures/89/89-003-009.html>).
- Remove the push rods and crossheads.
- Remove the rocker lever housing. Refer to Procedure 003-013 in Section 3 (</qs3/pubsys2/xml/en/procedures/89/89-003-013.html>).
- Remove the cylinder head. Refer to Procedure 002-004 in Section 2 (</qs3/pubsys2/xml/en/procedures/89/89-002-004.html>).
- Remove the piston cooling nozzles. Refer to Procedure 001-046 in Section 1 (</qs3/pubsys2/xml/en/procedures/89/89-001-046.html>).
- Remove the piston and connecting rod assembly and disassemble. Refer to Procedure 001-054 in Section 1 (</qs3/pubsys2/xml/en/procedures/89/89-001-054.html>).

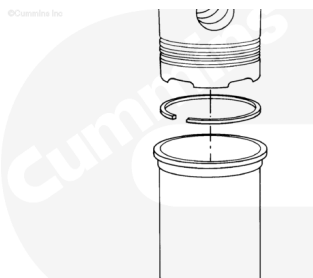
Remove

Use piston ring expander, Part Number 3164401, or equivalent, to remove the piston rings.



Measure

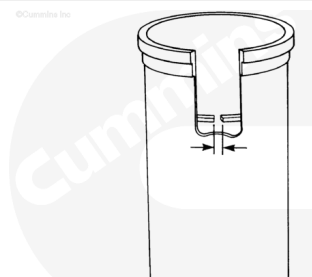
To check the ring gap, install the piston ring in the cylinder liner to be used. Use the top part of the piston to position the ring in the liner correctly.



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Use a feeler gauge to check the ring gap. The ring **must** be replaced if it does **not** meet specifications.

Piston Ring Gap			
	mm		in
Top	0.50	MIN	0.020
	0.65	MAX	0.026
Intermediate	0.70	MIN	0.028
	0.85	MAX	0.033
Oil	0.50	MIN	0.020
	0.70	MAX	0.028



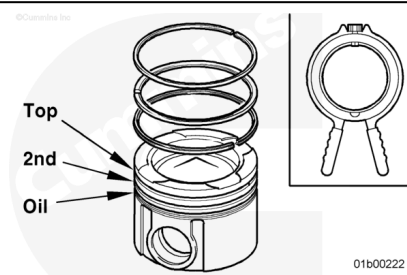
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Install

Note : Do not install the piston rings until the pistons are installed on the rods. Refer to Procedure 001-054 in Section 1. (</qs3/pubsys2/xml/en/procedures/89/89-001-054.html>)

Note : The top piston ring is marked with 1G or G3 depending on engine variation. The second piston ring is marked with 2G.

The markings on the rings **must** be positioned toward the top of the piston. There are no markings on the oil ring. The oil ring can be positioned either way. Use a ring expander, Part Number 3164401, or equivalent, and install the piston rings.

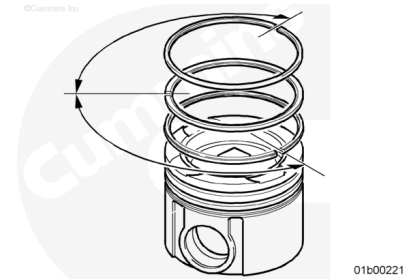


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Note : Do not align the gap of a ring with the piston pin bore.

Note : The gap in the oil ring expander must be turned 180 degrees opposite the gap on the oil ring.

Rotate the rings until the gaps are positioned as shown.



Finishing Steps

- Assemble and install the piston and connecting rod. Refer to Procedure 001-054 in Section 1 (</qs3/pubsys2/xml/en/procedures/89/89-001-054.html>).
- Install the piston cooling nozzles. Refer to Procedure 001-046 in Section 1 (</qs3/pubsys2/xml/en/procedures/89/89-001-046.html>).
- Install the cylinder head. Refer to Procedure 002-004 in Section 2 (</qs3/pubsys2/xml/en/procedures/89/89-002-004.html>).
- Install the rocker lever housing. Refer to Procedure 003-013 in Section 3 (</qs3/pubsys2/xml/en/procedures/89/89-003-013.html>).
- Install the push rods and crossheads.
- Install the rocker lever assembly. Refer to Procedure 003-009 in Section 3 (</qs3/pubsys2/xml/en/procedures/89/89-003-009.html>).
- Adjust the overhead set. Refer to Procedure 003-006 in Section 3 (</qs3/pubsys2/xml/en/procedures/89/89-003-006.html>).
- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3 (</qs3/pubsys2/xml/en/procedures/89/89-003-011.html>).
- Install the exhaust manifold. Refer to Procedure 011-007 in Section 11 (</qs3/pubsys2/xml/en/procedures/89/89-011-007.html>).
- Install the intake manifold. Refer to Procedure 010-023 in Section 10 (</qs3/pubsys2/xml/en/procedures/89/89-010-023.html>).
- Install the oil suction tube and block stiffener plate. Refer to Procedure 001-089 in Section 1 (</qs3/pubsys2/xml/en/procedures/89/89-001-089.html>).
- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7 (</qs3/pubsys2/xml/en/procedures/89/89-007-025.html>).
- Fill the engine with clean 15W-40 lubricating oil. Refer to Procedure 007-037 in Section 7 (</qs3/pubsys2/xml/en/procedures/89/89-007-037.html>) for engine oil capacity.
- Fill the cooling system. Refer to Procedure 008-018 in Section 8 (</qs3/pubsys2/xml/en/procedures/89/89-008-018.html>).
- Connect the battery or supply air to the starter.

